Gray means part of a previous submission (46 clubs, 17 hours)

Problem 1 (96IRNTST)

We claim the answer is $f \equiv 0$ or $f(x) = x^2$, which obviously work.

We plug in $y = \frac{f(x)-x^2}{2}$. We may get,

$$0 = 2f(x)(f(x) - x^2)$$

Therefore, for any x, either $f(x) \in \{0, x^2\}$.

Case 1: $f(x) \neq 0$ for any $x \neq 0$. This is the $f(x) = x^2$ case.

Case 2: $f(x) = 0$ for some $x \neq 0$.

We plug in such an x to get, for all y,

$$f(x^2 + y) = f(-y)$$

If $f(-y) = y^2$, it is impossible for $f(x^2 + y) \in \{x^4 + 2x^2y + y^2, 0\}$ to be equivalent to it unless if $x = 0$, which is impossible. Therefore, $f \equiv 0$ in this case.

Problem 2 (PEXIDER)

We claim that the answer is $f(x) \equiv kx + a + b; g(x) \equiv kx + a; h(x) \equiv kx + b$ for any real numbers a, b, k. This clearly works.

Plugging in $(x, 0)$, we have $f(x) = g(x) + h(0)$.

Plugging in $x = 0$, we have $f(y) = h(y) + g(0)$. Therefore,

$$\Leftrightarrow f(x + y) = f(x) + f(y) - h(0) - g(0)$$

We plug in $f'(x) = f(x) - h(0) - g(0)$ and use Cauchy's functional equation since f is continuous…

Problem 3 (08IMO4)

We claim the answer is $f(x) = x$ or $f(x) = \frac{1}{x}$. The latter obviously works. The former works since $wx = yz$.

Plug in $w = x = y = z = 1$,

$$\frac{2f(1)^2}{2f(1)} = 1 \Leftrightarrow f(1) = 1$$

We next plug in $y = w; z = x$ to get, for any positive w, x,

$$\frac{f(w)^2 + f(x)^2}{f(w^2) + f(x^2)} = 1 \Leftrightarrow f(w)^2 - f(w^2) = f(x^2) - f(x)^2$$

Plugging in $x = 1$, we get $f(w)^2 = f(w^2)$. Substituting this back in, for any positive $wx = yz$,

$$\frac{f(w^2) + f(x^2)}{f(y^2) + f(z^2)} = \frac{w^2 + x^2}{y^2 + z^2}$$

Since f is from positive reals to positive, we plug in positive numbers $a = w^2; b = x^2; c = y^2; d = z^2$ where $ab = cd$,

$$\frac{f(a) + f(b)}{f(c) + f(d)} = \frac{a + b}{c + d}$$

We now plug in $b = \frac{1}{a}; c = d = 1$.

$$f(a) + f\left(\frac{1}{a}\right) = a + \frac{1}{a} \quad (*)$$

$$\Leftrightarrow f(a^2) + 2f(a)f\left(\frac{1}{a}\right) + f\left(\frac{1}{a}\right)^2 = a^2 + 2 + \frac{1}{a^2}$$

From the other direction, we plug in $f(y)^2 = f(y^2);\ f(z)^2 = f(z^2)$. Therefore, for any positive $wx = yz$,

$$\frac{f(w)^2 + f(x)^2}{f(y)^2 + f(z)^2} = \frac{w^2 + x^2}{y^2 + z^2}$$

We plug in $y = z = 1$ and $x = \frac{1}{w}$.

$$f(w)^2 + f\left(\frac{1}{w}\right)^2 = w^2 + \frac{1}{w^2}$$

Combining this with the previous conclusion,

$$f(a)f\left(\frac{1}{a}\right) = 1$$

Plugging this back into $(*)$,

$$f(a) + \frac{1}{f(a)} = a + \frac{1}{a}$$

By bounding, this is possible only at $f(a) = a$ or $f(a) = \frac{1}{a}$, so $f(a) \in \left\{a, \frac{1}{a}\right\}$.

If $f(x) = x$ for some $x > 1$, then since f is continuous, $f(x) = x$ for all $x > 1$. We'd also get $f\left(\frac{1}{x}\right) = \frac{1}{x}$. Once again, since f is continuous, $f(x) = \frac{1}{x}$ for all $x < 1$ as well too, so $f(x) = x$.

Doing analogous work on $f(x) = \frac{1}{x}$, we have either $f(x) = x$ or $f(x) = \frac{1}{x}$ as desired. (the solution resolves the point-wise trap differently and I'm concerned whether this argument with continuous is valid)

Problem 4 (77IMO6)

We let $\{f(1), f(2), \ldots\} = \{a_1, a_2, \ldots\}$ be the multiset of outputs of f with $a_1 \leq a_2 \leq \cdots$.

Claim: $a_n = f(m) \Leftrightarrow n = m$

We prove this by induction.

Base case: $n = 1$.

FTSOC, if $a_1 = f(k)$ for some $k \neq 1$, then plug in $n = k - 1 \in \mathbb{Z}_{>0}$.

$$f(k) > f\big(f(k-1)\big)$$

This contradicts to $f(k)$ being the minimum. Therefore, $a_1 = f(1)$ and $a_1 \neq f(k)$ for any other positive integer k. This proves $a_1 = f(m) \Leftrightarrow n = 1$.

Inductive step: $n = k - 1 \rightarrow n = k$

FTSOC, let $a_k = f(x)$ for some $x \neq k$. By the inductive hypothesis, $x < k$ is impossible so $x > k$. We plug $n = x - 1$ to get,

$$f(x) > f\big(f(x-1)\big)$$

However, $f(x-1) \geq a_k$ by the induction hypothesis so $f\big(f(x-1)\big) \geq a_k$ as well. There is hence a contradiction, and we are done.

Conclusion: By mathematical induction, the claim is proven.

Since the claim gives us injectivity as well as $f(1) \leq f(2) \leq \cdots$, we have, f is strictly increasing. Therefore, $f(m) > f(n) \Leftrightarrow m > n$. Using this on our initial equation,

$$n + 1 > f(n)$$

$$\Leftrightarrow n \geq f(n)$$

Since $f: \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{>0}$, we quickly induct to get $f(n) = n$.

Problem 5 (17EGMO2)

We claim the answer is $k = 3$.

This can be achieved by coloring all the $n \equiv 0 \bmod 3$ one color, all the $n \equiv 1 \bmod 3$ another, and all the $n \equiv 2 \bmod 3$ yet another color. Our function f would be,

$$f(x) = \begin{cases} x, \text{if } x \equiv 1, 2 \bmod 3 \\ 1434x, \text{if } x \equiv 0 \bmod 3 \end{cases}$$

For two m, n of the same color, if $m \equiv n \equiv 1$ or $2 \bmod 3$, then $f(m) + f(n) = m + n = f(m + n)$. If $m \equiv n \equiv 0 \bmod 3$, then $f(m) + f(n) = 1434(m + n) = f(m + n)$. In this case, $m = 1; n = 2$ satisfies the second condition as $f(1 + 2) = 1434 \cdot 3 \neq 3$.

We now show the stronger claim that $k = 2$ is impossible (which shows $k = 1$ is impossible as well since you can choose one of the colors to fill everything with) even if the range of the function is all positive reals. Name the colors A, B.

Since the equalities are homogenous, we scale to let $f(1) = 1$ WLOG.

We seek to prove that if $k = 2$, $f(x) = x \; \forall \; x$.

Note that, since m has the same color as itself, $f(2m) = f(m) + f(m) = 2f(m)$. Therefore,

Lemma: $f(x) = x \Leftrightarrow f(2x) = 2x$

We now prove the claim using strong induction.

Base case: $x = 1$ is trivial.

Inductive step ($x = 1, \dots, k-1 \rightarrow x = k$)*:*

FTSOC, let $f(k) \neq k$.

Note that if k is even, then we've already proved $\frac{k}{2}$ so by the lemma $f(k) = k$. (!)

For odd k, we have already proved $\frac{k+1}{2}, \frac{k+3}{2}, \dots, \frac{2k-2}{2}$ so,

$$f(k+1) = k+1; f(k+3) = k+3; \dots; f(2k-2) = 2k-2$$

WLOG, let k be colored A. Note that pairs $(1, k-1), (2, k-2), \dots, \left(\frac{k-1}{2}, \frac{k+1}{2}\right)$ must have different colors since otherwise we may use (i) to get $f(k) = k$. In each of these pairs, if the odd element o is colored A, we have,

$$f(o) + f(k) = f(o+k)$$

Since $o + k$ is an even number less than or equal to $2k - 2$,

$$\Leftrightarrow o + f(k) = o + k \Leftrightarrow f(k) = k \quad (!)$$

Therefore, all the odd numbers less than k are colored B and all the evens are colored A. Therefore, we add to get $f(k+2) = f(k) + 2$. If $k + 2$ is colored B, then,

$$f(1) + f(k+2) = f(k+3)$$

$$\Leftrightarrow f(k) + 3 = k + 3 \quad (!)$$

However, if $k + 2$ is colored A,

$$f(k) + f(k+2) = f(2k+2)$$

$$\Leftrightarrow 2f(k) + 2 = 2(k+1) \quad (!)$$

Using proof by contradiction, the inductive step is complete.

Conclusion: By strong mathematical induction, we have proved that $f(k) \equiv k$.

If $f(x) = x$, then we cannot have $f(m+n) = f(m) + f(n)$ so $k = 2$ does not work.

Problem 6 (15SLA2)

The answers are $f(x) \equiv x+1$ or $f \equiv -1$. Both obviously work.

Plugging in $(x, f(x))$, we have,

$$f\left(x - f(f(x))\right) = -1$$

We let $t = x - f(f(x))$. Then, plugging in $y = t$,

$$f(x+1) = f(f(x))$$

Plugging in $(f(x), x)$,

$$f(0) = f\left(f(f(x))\right) - f(x) - 1$$

$$\Leftrightarrow f(0) + 1 = f(x+2) - f(x)$$

Therefore, for some integer $c = \frac{f(0)+1}{2}, a, b$, [citation needed?]

$$f(x) \equiv \begin{cases} cx + a \text{ if } x \text{ is even} \\ cx + b \text{ if } x \text{ is odd} \end{cases}$$

Plugging in $x = 0$ gives $a = 2c - 1$.

$$f(x) \equiv \begin{cases} c(x+2) - 1 \text{ if } x \text{ is even} \\ cx + b \text{ if } x \text{ is odd} \end{cases}$$

Case 1: c is even.

Sketch: Plugging in we get contradiction if $c \neq 0$. Plugging $c = 0$ back, we get $b = -1$ so $f \equiv -1$.

Case 2: c is odd.

Sketch: Plugging in we get contradiction if $c \neq 1$. Plugging $c = 1$ back, we get $b = 1$ so $f \equiv x + 1$.

+2.5 hours

Problem 7 (EGMO 2021/2)

We claim that the solutions are $f(x) \equiv x$ or $f(x) \equiv -x$. These

Lemma: $a \neq 0 \Longrightarrow f(a) \neq 0$

FTSOC, $a \neq 0$ and $f(a) = 0$, we plug in $x = a; y = 0$,

$$f(a\,f(a)) = f(0) + a^2$$

$$\Leftrightarrow f(0) = f(0) + a^2 \Leftrightarrow a = 0 \quad (!)$$

If we plug in $y = kxf(x)$ for some integer k,

$$f\big((k+1)x\,f(x)\big) = f\big(kx\,f(x)\big) + x^2$$

Therefore, by induction, for any quotients x, y and integer n,

$$f\big(n \cdot x\,f(x)\big) = f(0) + n \cdot x^2$$

Since $x\,f(x)$ is a quotient, for any non-zero quotients a, b, we can choose non-zero integers m, n such that,

$$m \cdot a\,f(a) = n \cdot b\,f(b)$$

$$\Longrightarrow f\big(m \cdot a\,f(a)\big) = f\big(n \cdot b\,f(b)\big)$$

$$\Longleftrightarrow f(0) + m \cdot a^2 = f(0) + n \cdot b^2$$

$$\Longleftrightarrow ma^2 = nb^2$$

Since m, a are non-zero, $f(a)$ is non-zero as well. $m \cdot a\,f(a) = n \cdot b\,f(b) \neq 0$. We can hence divide the last equation by the first to get, for any non-zero quotients a, b.

$$\frac{ma^2}{m \cdot a\,f(a)} = \frac{nb^2}{m \cdot b\,f(b)}$$

$$\Longleftrightarrow \frac{a}{f(a)} = \frac{b}{f(b)}$$

Fixing a non-zero b, we have $\frac{a}{f(a)} = k$ for a constant quotient k. Therefore,

$$f(x) \equiv kx$$

We plug this in,

$$f(kx^2 + y) = ky + x^2$$

Plugging in $y = 0$,

$$\Longrightarrow k^2x^2 = x^2$$

Therefore, $k = \pm 1$ and we are done.

Problem 8 (11TSTST1)

We claim that either $f(a,b) \equiv a$ or $f(a,b) \equiv b$.

Plugging in $a = b = c$, the median of $f(a,a), f(a,a), f(a,a)$ is a. Therefore,

$$f(a,a) = a$$

Now plugging in $b = a \neq c$, plugging in,

$$\mathrm{median}\big(f(a,b), f(b,c), f(c,a)\big) = \mathrm{median}(a,b,c)$$

$$\Longleftrightarrow \mathrm{median}\big(f(a,a), f(a,c), f(c,a)\big) = a$$

Since $f(a,a) = a$ already, for any distinct integers a, c,

$$a \in \{f(a,c), f(c,a)\}$$

By symmetry, we also have,

$$c \in \{f(a,c), f(c,a)\}$$

Therefore, we get, $f(a,b) \in \{a,b\}$ and $f(b,a)$ is the other selection.

We resolve the point-wise trap using proof by contradiction.

FTSOC, let, for numbers $a \neq b; c \neq d$,

$$f(a,b) = a; f(c,d) = d$$

If $a > b$, we may switch the positions of a, b to get $f(b,a) = b$. WLOG, let $a < b$ and $c < d$.

Claim: $f(a,b) = a \Longrightarrow f(a,d) = a$ whenever $a < b, d$ or $a > b, d$.

FTSOC, if $f(a,d) = d$ in this case,

$$f(a,b) = a; f(a,d) = d \Leftrightarrow f(a,b) = a;\ f(d,a) = a$$

Therefore, the median of $f(a,b), f(b,d), f(d,a)$ is a. However, we are given that the median of a, b, d is not a. (!)

We immediately have $a = c$ is impossible. WLOG, let $a < c$. Since $d > c$ and $d > c > a$,

$$f(c,d) = d \Leftrightarrow f(d,c) = c \Longrightarrow f(d,a) = c$$

Since $c \neq a; c \neq d$, we have a contradiction, and the point-wise trap is resolved.

Problem 9 (20ELMO1)

This solution is much simpler than the official solution, which makes me think I did something wrong.

We claim that $f(x) \equiv x + 1$, which works because $f^{f^{f(x)}(y)}(z) = f^{f^{x+1}(y)}(z) = x + y + z + 1$.

Plugging in $(x, y, f(z))$,

$$f^{f^{f(x)}(y)+1}(z) = x + y + f(z) + 1$$

Taking f of the original equation,

$$f^{f^{f(x)}(y)+1}(z) = f(x + y + z + 1)$$

Therefore,

$$f(z+3) = f(z) + 3;$$

$$f(z+4) = f(z) + 4$$

By induction, $f(z) \equiv (z-1) + f(1)$. We write $f(z) \equiv z + c$ for some c. Plugging this in, we clearly get $c = 1$.

Problem 10 (18TSTST1)

We claim the solution is $\theta(p) \equiv p(n)$ for any integer n. This clearly works as $p(n) \mid p(n)q(n)$.

For $p(x) = x$, we let $\theta(p) = n$ for some integer n. We prove that $\theta(p) = p(n)$ for all polynomials p using induction on the degree of p.

Base case: $\deg(p) = 0$

We plug in $p(x) = a$.

$$\theta(p+1) = \theta(p) + 1 \Leftrightarrow \theta(a+1) = \theta(a) + 1$$

By induction, we may write, for any integer a, $\theta(a) \equiv a + k$ for some constant k. Plugging in $p(x) = a$ and $q(x) = 0$, $a + k \mid k$. This is true for all integers a, so anything divides k meaning $k = 0$. For degree zero $p(x) = a$, $\theta(p) = a$ as desired.

Base case 2: $\deg(p) = 1$

We have $\theta(x+b) = n + b$ for any integer b. Therefore, for all integers $a, k \neq -c$,

$$\theta(x+k) \mid \theta(ax+ak)$$

$$\Longrightarrow n + k \mid \theta(ax) + ak$$

$$\Leftrightarrow n + k \mid \theta(ax) - an$$

For a sufficiently large k, we have $\theta(an) = an$ for the RHS to be zero. Therefore, $\theta(ax+b) = an + b$.

Strong inductive step: $\deg(p) = k \; for \; k \geq 2$

For any polynomial p and integers a, b, we have $a - b \mid p(a) - p(b)$.

For any integer m for $m \neq -n$, note that,

$$\theta\left(\frac{p - p(m)}{x - m}\right) \mid \theta\big(p - p(n)\big)$$

$$\Leftrightarrow \frac{p(n) - p(m)}{n - m} \mid \theta(p) - p(n)$$

$$\Leftrightarrow \frac{p(n) - p(m)}{n - m} \mid \theta(p) - p(m)$$

As $\deg(p) \geq 2$, as $n \to \infty$, the LHS grows increasingly greater while the RHS remains fixed. Therefore,

$$\theta(p) = p(m)$$

Problem 11 (19USMCA4)

(This solution does not use the injectivity argument :D)

We claim the answer is $f(x) \equiv x + c$ or $f(x) \equiv 0$, for some real number c. The latter case obviously works, and the former works since, we'd have,

$$(x + 2c + y)^2 = (x - y)(x - y) + 4(x + c)(y + c)$$

If we expand, we find that this is true. We now prove these are the only solutions.

We plug in $y = -f(x) + x$ to get,

$$f(x)^2 = f(x)\big(f(x) - f(-f(x) + x)\big) + 4f(x)\, f(-f(x) + x)$$

Therefore, for all x,

Option 1: $f(x) = 0$.

Option 2: If $f(x) \neq 0$, we have,

$$f(x) = f(x) - f(-f(x) + x) + 4\, f(-f(x) + x)$$

$$\Leftrightarrow f(-f(x) + x) = 0$$

Case 1: $f(0) \neq 0$

$$f(-f(0) + 0) = f\big(-f(0)\big) = 0$$

In that case, plugging in $x = -f(0)$, we have, for all y,

$$f(y)^2 = (-f(0) - y)\big(-f(y)\big)$$

We have either $f(y) = 0$ or, if not, $f(y) = y + f(0)$. In summary, for all real x,

$$f(x) \in \{0, x + f(0)\}$$

We claim that either $f \equiv 0$ or $f \equiv x + f(0)$. FTSOC, let there exists two real numbers m, n such that $m \neq n$ and $m, n \neq -f(0)$ such that $f(m) = 0$ but $f(n) = n + f(0)$. We plug in $x = m$, $y = n$ to get,

$$f(n) = (m - n)\big(-n - f(0)\big)$$

$$\Leftrightarrow n + f(0) = (m - n)\big(-n - f(0)\big)$$

Since $n \neq -f(0)$,

$$\Leftrightarrow n - m = 1$$

This means that all such pair m, n must satisfy $n - m = 1$. Now consider $f\left(\frac{m+n}{2}\right)$. No matter the choice, we'd have a pair with difference other than one.

Case 2: $f(0) = 0$

We claim that $f \equiv 0$ or $f(x) \equiv x$. We plug in $x = 0$,

$$f(y)^2 = -y\big(-f(y)\big)$$

Therefore, $f(x) \in \{0, x\}$ for all x. FTSOC, let there be a pair of two distinct non-zero real numbers m, n such that $f(m) = m$ but $f(n) = 0$. We plug in $x = m; y = n$ to get,

$$f(m+n)^2 = (m-n)m$$

Since $m \neq n$ and $m \neq 0$, $f(m+n) \neq 0$ meaning $f(m+n) = m+n$. Therefore,

$$(m+n)^2 = (m-n)m$$

$$\Leftrightarrow m^2 + 2mn + n^2 = m^2 - mn$$

$$\Leftrightarrow 3mn + n^2 = 0$$

Since $n \neq 0$, $m = -\frac{1}{3}n$ for all such pairs m, n. Now consider $f\left(\frac{1}{2}n\right)$. If $f\left(\frac{1}{2}n\right) = 0$, then we still have $f\left(-\frac{1}{3}n\right) = -\frac{1}{3}n$ but $-\frac{1}{3}n \neq -\frac{1}{3}\left(\frac{1}{2}n\right)$. Similarly, if $f\left(\frac{1}{2}n\right) = \frac{1}{2}n$, we still have $f(n) = 0$ but $\frac{1}{2}n \neq -\frac{1}{3}(n)$ so both leads to contradiction. Therefore, it is impossible for there to be such a pair.

The point-wise trap is resolved, and we have our answers.

Problem 12 (07MOPFE)

We claim the answer is $f(x) = \frac{1}{x}$, which obviously works.

We show that for positive real numbers x, y.

$$f(x)y + \frac{x}{y} = x^2 + y^2$$

$$\Leftrightarrow 0 = y^3 - f(x)y^2 + x^2y - x$$

We let $g(y) = y^3 - f(x)y^2 + x^2y - x$ and show that g has a real positive root. Since g is continuous $\lim_{y \to 0^+} g(y) = -x$ and $\lim_{y \to +\infty} g(y) = +\infty$ noting that x is non-zero so $-x$ is negative, by the Intermediate Value Theorem, there must be a real positive root. (The IVT is usually applied on closed intervals so idk if this is valid)

We may plug such a y in to get, for some $t = f(x)y + \frac{x}{y} = x^2 + y^2$,

$$f(t) = xy \cdot f(t)$$

Since f is positive,

$$\Leftrightarrow 1 = xy$$

Plugging $y = \frac{1}{x}$ back into, $y^3 - f(x)y^2 + x^2y - x$,

$$0 = \frac{1}{x^3} - \frac{f(x)}{x^2} + x - x$$

$$\Leftrightarrow 0 = 1 - x\,f(x)$$

$$\Leftrightarrow f(x) = \frac{1}{x}$$

This is as desired.

+5.5 hours

Problem 13 (20RMM4)

Since f is surjective, we may define let g be any function that $g(x)$ satisfies $f\big(g(x)\big) = x$. We immediately have that g is injective.

If a set B is not sum-free, note that $\{g(b): b \in B\}$ is also not sum-free for if it was, then plugging in these values in $\{g(b): b \in B\}$ into f will give $\{f\big(g(b)\big): b \in B\} = B$ is also sum-free, which it isn't (!).

Main Claim: $g(x) = x \cdot g(1)$

Claim: $f(2x) = 2 \cdot f(x)$

Taking $B = \{x, 2x\}$, we get either $f(2x) \in \left\{2 \cdot f(x), \frac{1}{2} f(x)\right\}$. In the latter case, we'd induct to get $f\big(2^k x\big) = \frac{x}{2^k}$. For sufficiently large k contradiction due to this being not an integer.

We now induct to prove the main claim.

Base case: $x = 1$. Trivial.

Strong Inductive Step: $x = 1, \dots, k-1 \to x = k$

If k is even, then by claim we are done. Otherwise, plug in $B = \{1, k, k+1\}$. Since $k+1$ is even, by the inductive hypothesis, $f(k+1) = 2\,f\left(\frac{k+1}{2}\right) = (k+1)\,f(1)$.

The image of B is $\{f(1), f(k), (k+1)\,f(k)\}$. Use injective to get more contradictions… $f(k) = k\,f(1)$.

Conclusion: By induction $f(x) \equiv x \cdot f(1)$.

With the claim proven, note that f is injective, since if $a \neq b$ but $f(a) = f(b)$, $g\big(f(a)\big)$ can be either a or b without changing any other values of g. This contradicts g being linear. Therefore, since f is injective, we have $f(x) \equiv x$.

Problem 14 (14TWNQ1J4)

We claim that $f(x) \equiv x^3 - 1$, which obviously works since it is strictly increasing and,

$$m^3 n^3 - 1 = m^3 - 1 + n^3 - 1 + (m^3 - 1)(n^3 - 1)$$

We plug in $g(x) \coloneqq f(x) + 1 \Leftrightarrow f(x) = g(x) - 1$,

$$g(mn) - 1 = g(m) - 1 + g(n) - 1 + g(m)g(n) - g(m) - g(n) + 1$$

$$\Leftrightarrow g(mn) = g(m)g(n)$$

Additionally, $g(2) = 8$. We claim that $g(x) \equiv x^3$, which would be sufficient.

Plugging in $m = 1; n = 2$, we have $g(1) = 1$. Plugging in $m = 0; n = 2$, we have $g(0) = 0$.

We can induct for the rest using bounding using the strictly increasing…

Problem 15 (21CAMO2)

We claim that the answer is, for any set S closed under multiplication and division,

$$f(x) = \begin{cases} 0 \text{ if } x \notin S \\ \sqrt{x} \text{ if } x \in S \end{cases}$$

We may verify that this works. For any pair x, y,

Case 1: $f(x) = 0; f(y) = 0$. Trivially both sides are zero.

Case 2: $f(x) = 0; f(y) = \sqrt{y}$

Plugging in, we indeed have,

$$\left(\sqrt{y}\right)^3 = y\sqrt{y} + 3f(xy)\left(0 + \sqrt{y}\right)$$

However, since S is division, if $f(xy) = \sqrt{xy}$ and $f(y) = \sqrt{y}$ then $f(x) = \sqrt{x}$, which is false. Therefore, $f(xy) = 0$ and this is clearly true.

The case for $f(x) = \sqrt{x}; f(y) = 0$ is analogous.

Case 3: $f(x) = \sqrt{x}; f(y) = \sqrt{y}$

We have $f(xy) = \sqrt{xy}$. We plug in to get this is true.

We plug in $x = y = 1$ to get,

$$8f(1)^3 = 2f(1) + 6f(1)^2$$

The non-negative solutions to this are $f(1) = 0$ and $f(1) = 1$.

Case 1: $f(1) = 0$

We claim this implies $f \equiv 0$. FTSOC, if there exists a y that gives $f(y) \neq 0$,

We plug in $x = 1$ to get,

$$f(y)^3 = y\, f(y) + 3\, f(y)^2$$

$$\Leftrightarrow f(y)^2 - 3f(y) - y = 0$$

Since $y > 0$, the only positive solution for $f(y) = \frac{3+\sqrt{9+4y}}{2}$ by the quadratic formula. However, plugging in $x = y$ into the original equation for this y,

$$\left(3+\sqrt{9+4x}\right)^3 = x\left(3+\sqrt{9+4x}\right) + 3f(x^2)\left(3+\sqrt{9+4x}\right)$$

Obviously, if $f(x^2) = 0$, then by bounding this is impossible. Therefore, $f(x^2) \neq 0 \Longrightarrow f(x^2) = \frac{3+\sqrt{9+4x^2}}{2}$. Plugging in,

$$\left(3+\sqrt{9+4x}\right)^3 = x\left(3+\sqrt{9+4x}\right) + 3\frac{\left(3+\sqrt{9+4x^2}\right)}{2}\left(3+\sqrt{9+4x}\right)$$

$$\Leftrightarrow \left(3+\sqrt{9+4x}\right)^2 = x + 3\frac{\left(3+\sqrt{9+4x^2}\right)}{2}$$

Once again, by bounding, there are no positive solutions.

This gives $f \equiv 0$ one of our solutions.

Case 2: $f(1) = 1$

We plug in $x = 1$ to get,

$$\left(1+f(y)\right)^3 = 1 + yf(y) + 3f(y)\left(1+f(y)\right)$$

$$\Leftrightarrow f(y)^3 + 3f(y)^2 + 3f(y) = yf(y) + 3f(y) + 3f(y)^2$$

$$\Leftrightarrow f(y)^3 = yf(y)$$

Therefore, the non-negative solutions are either $f(y) = 0$ or $f(y) = \sqrt{y}$.

However, if $f(x) = \sqrt{x}$, $f(y) = \sqrt{y}$, we have, plugging in,

$$\left(\sqrt{x}+\sqrt{y}\right)^3 + x\sqrt{x} + y\sqrt{y} + 3f(xy)\left(\sqrt{x}+\sqrt{y}\right)$$

$$\Leftrightarrow f(xy) = \sqrt{xy}$$

Therefore, S is a multiplicative subgroup.

Problem 16 (18TURTST2)

We claim that $f(x) \equiv 2x$, which obviously works.

We plug in $x = f(y) - 2y$ to get, for all real y,

$$f(f^2(y) - 2yf(y) + y^2) = f((f(y) - y)^2) - (f(y) - 2y)f(f(y) - 2y)$$

$$\Leftrightarrow (f(y) - 2y) \cdot f(f(y) - 2y) = 0 \qquad (*)$$

Taking $y = 0$, we see that either $f(0) = 0$ or $f\left(f(0)\right) = 0$. However, $f(0) = 0$ implies $f\left(f(0)\right) = 0$ anyways so $f\left(f(0)\right) = 0$ is always true.

Claim: $f(a) = 0 \Longrightarrow a = f(0)$.

FTSOC, if $a \neq 0$ and $f(a) = 0$, we plug in $(x, y) = \big(a, f(0)\big)$ to get,

$$f(f(0)^2) = f\left(\big(a + f(0)\big)^2\right)$$

Plugging in $(x, y) = (f(0), a)$,

$$f(a^2) = f((f(0) + a)^2)$$

Clearly,

$$f(f(0)^2) = f(a^2)$$

We now plug in $y = a$ into the original equation,

$$f(a^2) = f((x + a)^2) - x\, f(x)$$

Similarly, plugging in $y = f(0)$ gives,

$$f(f(0)^2) = f\left(\big(x + f(0)\big)^2\right) - x\, f(x)$$

This means $f((x + a)^2) = f\left(\big(x + f(0)\big)^2\right)$.

Next, we plug in, $(x, y) = \big(x + a - f(0), f(0)\big)$,

$$f(f(0)^2) = f((x + a)^2) - \big(x + a - f(0)\big) f\big(x + a - f(0)\big)$$

$$\Leftrightarrow\ f\left(\big(x + f(0)\big)^2\right) - x\, f(x) =\ f\left(\big(x + f(0)\big)^2\right) - \big(x + a - f(0)\big) f\big(x + a - f(0)\big)$$

We let $c \coloneqq a - f(0)$,

$$x\, f(x) = (x + c)\, f(x + c)$$

We now plug in $(x, y) = (x + c, y)$ into the original equation,

$$f\big((x + c)\, f(y) + y^2\big) = f((x + y + c)^2) - (x + c)\, f(x + c)$$

Combining this with the first equation, since $f((x + y + c)^2) = f((x + y)^2)$,

$$f\big((x + c)\, f(y) + y^2\big) = f(x\, f(y) + y^2)$$

As f is surjective, we plug in the selection $y = k$ such that $f(y) = 1$,

$$f(x + c + k^2) = f(x + k^2)$$

Since x can be any real number,

$$\Leftrightarrow f(x + c) = f(x)$$

However, using $(x + c)\, f(x + c) = x\, f(x)$, we have $x + c = x \Leftrightarrow c = 0$. (!) Therefore we are done.

Claim: $f(0) = 0$

Plugging in $y = f(0)$ into $(*)$,

$$\left(f\big(f(0)\big) - 2f(0)\right) \cdot f\left(f\big(f(0)\big) - 2f(0)\right) = 0$$

FTSOC, if $f(0) \neq 0$, then the first factor is non-zero and,

$$f\big(-2f(0)\big) = 0$$

$$\Longrightarrow -2f(0) = f(0)$$

Therefore, $f(0) = 0$. (!)

Claim: $f(x) = 2x$

We simply use $(*)$, FTSOC, if there exists a quantity a such that $f(a) \neq 2a$,

$$(f(a) - 2a) \cdot f(f(a) - 2a) = 0$$

$$\Leftrightarrow \ f(f(a) - 2a) = 0$$

$$\Leftrightarrow f(a) = 2a \quad (!)$$

We have proved what was necessary.

+8 hours

Problem 17 (14AMO2)

The solutions are $f(x) \equiv 0$ or $f(x) \equiv x^2$. Both can be checked to work.

Claim: $f(0) = 0$.

FTSOC, if $f(0) \neq 0$, we plug in $x = f(0)$,

$$f(0)\, f\big(2f(y) - f(0)\big) + y^2\, f\big(2f(0) - f(y)\big) = \frac{f\big(f(0)\big)^2}{f(0)} + f\big(y\, f(y)\big)$$

Plugging in $y = 0$,

$$f(0)\, f\big(f(0)\big) = \frac{f\big(f(0)\big)^2}{f(0)} + f(0)$$

$$\Leftrightarrow f(0)^2\Big(f\big(f(0)\big) - 1\Big) = f\big(f(0)\big)^2$$

$$\Leftrightarrow f(0) = \pm \frac{f\big(f(0)\big)}{\sqrt{f\big(f(0)\big) - 1}}$$

Since $f(0)$ is an integer, $\frac{f(f(0))}{\sqrt{f(f(0))-1}}$ is an integer. Note that $f\big(f(0)\big)$ is an integer too. Therefore,

$$\sqrt{f(f(0)) - 1} \mid f(f(0))$$

$$\Leftrightarrow f(f(0)) - 1 \mid f(f(0))^2$$

$$\Leftrightarrow \; f(f(0))^2 \equiv 0 \bmod f(f(0)) - 1$$

$$\Leftrightarrow \; 1 \equiv 0 \bmod f(f(0)) - 1$$

This is clearly impossible if $f(f(0)) > 2$. Since $f(f(0)) - 1$ is positive, we must have $f(f(0)) = 2$. However, then, $f(0) = \pm 2$.

Plugging in $x = 3; y = 0$,

$$3\, f(4 - x) = \frac{f(3)^2}{3} + f(0)$$

If $f(3)$ is not a multiple of 3, then $\frac{f(3)^2}{3}$ is a fraction (!). Therefore, $3 \mid f(3)$. Taking mod 3, $3 \mid f(0)$ as well which contradicts $f(0) = \pm 2$. (Here I noticed I took a longer than necessary route)

We now plug in $y = 0$ to get, for all $x \neq 0$,

$$x\, f(-x) = \frac{f(x)^2}{x}$$

$$\Leftrightarrow x^2\, f(-x) = f(x)^2$$

Plugging in $-x$ into this equation,

$$x^2\, f(x) = f(-x)^2$$

$$\Leftrightarrow x^2\, f(x) = \left(\frac{f(x)^2}{x^2}\right)^2$$

$$\Leftrightarrow x^6\, f(x) = f(x)^4$$

Therefore, $f(x) \in \{0, x^2\}$ for all x. We now prove that $f \equiv 0$. Plugging in $f(x) \equiv x^2$ into the original equation, we see that it does not work. Therefore, there must exist some $m \neq 0$ such that $f(m) = 0$.

Consider the case where there's a non-zero m such that $f(m) = 0$, we prove that $f \equiv 0$ in that case. We plug in $y = n$ to get,

$$x\, f(-x) + n^2 f(2x) = \frac{f(x)^2}{x} + f(0)$$

However, note that $x^2 f(-x) = f(x)^2$ and $f(0) = 0$, therefore,

$$n^2 f(2x) = 0$$

$$\Leftrightarrow f(2x) = 0$$

Therefore, we are done with all the evens. Plugging in $x = 2k$, for all y,

$$2k\, f\big(2(f(y)-k)\big) + y^2\, f\big(4k - f(y)\big) = \frac{f(2k)^2}{2k} + f\Big(y\big(f(y)\big)\Big)$$

$$\Leftrightarrow y^2\, f\big(4k - f(y)\big) = f\Big(y\big(f(y)\big)\Big)$$

FTSOC, if $f(y) = y^2$ for some non-zero y,

$$\Leftrightarrow y^2\, f(4k - y^2) = f(y^3)$$

Clearly, this cannot be true if $f(y^3) \neq 0 \Leftrightarrow f(4k - y^2) \neq 0$ for a sufficiently large k by bounding. Therefore, $f(y^3) = 0$ for all y and hence $f(4k - y^2) = 0 \Leftrightarrow f(y^2 - 4k) = 0$.

The former covers all the $3 \bmod 4$ numbers except for $-y^2$, and the latter covers all the $1 \bmod 4$ except y^2. Therefore, we have $f(x) = 0$ for all $x \neq \pm y^2$.

Since $f(y) = y^2$, we have $y = \pm y^2$. Since y is non-zero, $y = \pm 1$ are the only options. Note that,

$$f(1) \neq 0 \Leftrightarrow f(-1) \neq 0$$

We may hence look at whether the function,

$$f(x) = \begin{cases} 1, \text{if } x = \pm 1 \\ 0, \text{otherwise} \end{cases}$$

May exist. Plugging in $x = -1; y = 1$,

$$-1 \cdot f(3) + f(-3) = 1 + 1$$

$$\Leftrightarrow 0 = 2$$

This does not work.

The point-wise trap is hence resolved. $f(x) \equiv 0$ or $f(x) \equiv x^2$.

Problem 18 (12AMO4)

We claim the solutions are $f \equiv 1, f \equiv 2, f(x) \equiv x$. These obviously work.

Claim: $f(n) \geq n$ or $f(n) \in \{1, 2\}$

FTSOC, if not, plug in $m = n!$ to get,

$$n! - n \mid f(n!) - f(n) \Leftrightarrow n! - n \mid f(n)! - f(n)$$

By bounding, this is clearly impossible for $n > 2$ and $f(n) > 2$.

Case 1: $f(n) = 1$ for some $n > 2$

We claim that $f \equiv 1$. FTSOC, if there exists $f(m) = k \neq 1$, plug in m to get,

$$m - n \mid k - f(n)\ \forall\, n \neq m$$

Note that if $f(n) = 1$, we have $f(n!) = f(n)! = 1$ and so on. Therefore, we may plug in a sufficiently large value of n on the LHS and still have $f(n) = 1$. Therefore, $k = 1$.

Case 2: $f(n) = 2$ for some $n > 2$

This case is analogous, we have $f \equiv 2$.

Case 3: $f(n) \geq n$ for some $n > 2$

We use induction to prove that $f(x) \equiv x$.

Claim: $f(2) = 2$

Note that $f(2) \in \{1, 2\}$. FTSOC, let $f(2) = 1$, we have, for any $m > 2$,

$$m! - 2 \mid f(m)! - 1$$

Since $f(m) \geq m$, the RHS is odd, but the LHS is even. (!)

Claim: $f(1) = 1$

FTSOC, if $f(1) = 2$, we plug in $m = 3!;\ n = 1$ into the original equation to get,

$$3! - 1 \mid f(3)! - 2$$

Since $f(3)! - 2 \equiv 0 \bmod 5$, clearly $f(3) \geq 5$ doesn't work. We may check to find $f(3) = 2$ is the only solution. (!)

Claim: $f(3) = 3$

We plug in $m = 3!; n = 1$ into the original equation to now get,

$$3! - 1 \mid f(3)! - 1$$

By a similar process, we may deduct $f(3) = 3$.

We now finally prove that $f(x) \equiv x$. FTSOC, if there exists a $n > 3$ such that $f(n) \neq n$, we plug in such a n to have,

$$m - n \mid f(m) - f(n)\ \forall\ m \neq n$$

Note that, $f(3!) = 3!; f\left((3!)!\right) = (3!)!$ and so on. We plug in a sufficiently large m in the pattern to get, by bounding, $f(n) = n$.

<u>Problem 19 (04IMO2)</u>

We claim the answer is $P(x) = ax^4 + bx^2$ for some real numbers a, b, which can be verified to work…

Plugging in $(0, 0, 0)$,

$$3P(0) = 2P(0) \Leftrightarrow P(0) = 0$$

Plugging in $(2x, 2x, -x)$ for any real x, noting that $(2x)(2x) + (2x)(-x) + (-x)(2x) = 0$,

$$P(0) + P(3x) + P(-3x) = 2P(3x)$$

$$\Leftrightarrow P(3x) = P(-3x)$$

Therefore, P is even. We now prove that $\deg(P) < 6$ to finish.

Plugging in $(-2x, 3x, 6x)$, noting that $(-2x)(3x) + (3x)(6x) + (6x)(-2x) = 0$,

$$P(-5x) + P(-3x) + P(8x) = 2P(7x)$$

$$\Rightarrow \lim_{x\to\infty} P(-5x) + P(-3x) + P(8x) = \lim_{x\to\infty} 2P(7x)$$

If $m \coloneqq \deg(P) \geq 6$, then,

$$\Leftrightarrow (-5x)^m + (-3x)^m + (8x)^m = 2 \cdot (7x)^m$$

Note that m is even and $x \neq 0$ since we are taking it large,

$$\Leftrightarrow 5^m + 3^m + 8^m = 2 \cdot 7^m$$

For $m \geq 6$, $8^m > 2 \cdot 7^m$ so we have a contradiction.

+11 hours

Problem 20 (13IMO5)

Claim 1: $f(x) \geq x$

We let $g(a) = f(a) - a$,

$$f(x)f(y) \geq f(xy) \Leftrightarrow x\,g(y) + y\,g(x) \geq 0;$$

There exists a rational $a > 1$ with $f(a) = a \Leftrightarrow$ There exists a rational $a > 1$ with $g(a) = 0$.

We plug in $x = a$ to get, for all y in the domain of f,

$$a\,g(y) \geq 0 \Leftrightarrow g(y) \geq 0 \Leftrightarrow f(y) \geq y$$

Claim 2: If $f(t) = t$, then $f(r) = r$ for all $r \leq t$.

Note that,

$$f(x+y) \geq f(x) + f(y) \Leftrightarrow g(x+y) \geq g(x) + g(y);$$

Plugging in $x = r; y = t - r$ we are done using claim 1.

Applying (i) repeatedly, we have, using claim 1,

$$f(a)^k = a^k \geq f\left(a^k\right) \Rightarrow f\left(a^k\right) = a^k \ \forall \text{ integer } k$$

Therefore, using claim 2, as $a > 1$, we may get $f(x) \equiv x$ as desired.

Problem 21 (16SLA4)

Claim: $f(1) = 1$

Plug in $x = y = 1$,

$$f(1)f\big(f(1)\big) + f\big(f(1)\big) = f(1)f\big(f(1)\big) + f(1)f\big(f(1)\big)$$

$$\Leftrightarrow f\big(f(1)\big) = f(1)f\big(f(1)\big)$$

Since f is non-zero, we are done. Plugging in $x = 1$,

$$f\big(f(y)\big) + f(y) = f(y)\left(1 + f\big(f(y^2)\big)\right)$$

$$\Leftrightarrow f\big(f(y)\big) = f(y) \cdot f\big(f(y^2)\big) \quad (A)$$

However, plugging in $y = 1$,

$$x\, f(x^2) + f\big(f(x)\big) = f(x)\big(f\big(f(x^2)\big) + 1\big)$$

$$\Leftrightarrow x\, f(x^2) + f\big(f(x)\big) = f\big(f(x)\big) + f(x)$$

$$\Leftrightarrow x\, f(x^2) = f(x) \quad (B)$$

Claim: f is injective

FTSOC, if $f(a) = f(b)$ but $a \neq b$,

For any positive real c, we plug in $(x, y) = (a, c)$,

$$a\, f(a^2)\, f\big(f(c)\big) + f\big(c\, f(a)\big) = f(ac)\left(f\big(f(a^2)\big) + f\big(f(c^2)\big)\right)$$

$$\Leftrightarrow f(a)\, f\big(f(c)\big) + f\big(c\, f(a)\big) = f(ac)\left(\frac{f\big(f(a)\big)}{f(a)} + \frac{f\big(f(c)\big)}{f(c)}\right)$$

Plugging in $(x, y) = (b, c)$,

$$f(b)\, f\big(f(c)\big) + f\big(c\, f(b)\big) = f(bc)\left(\frac{f\big(f(b)\big)}{f(b)} + \frac{f\big(f(c)\big)}{f(c)}\right)$$

Substituting $f(a) = f(b) \Longrightarrow f\big(f(a)\big) = f\big(f(b)\big)$ and dividing,

$$f(ac) = f(bc)\ \forall\ c \in (0, \infty)$$

We may plug in $c = f(a) = f(b)$ to get,

$$f\big(a\, f(a)\big) = f\big(b\, f(b)\big)$$

Plugging in $(x, y) = (a, a)$, using (A) and (B).

$$\Leftrightarrow f(a)\, f\big(f(a)\big) + f\big(a\, f(a)\big) = 2f(a^2)f\big(f(a^2)\big)$$

$$\Leftrightarrow af(a)f\big(f(a)\big) + af\big(af(a)\big) = 2f(a)f\big(f(a^2)\big)$$

$$\Leftrightarrow\ a\left(f(a)f\big(f(a)\big) + f\big(af(a)\big)\right) = 2f\big(f(a)\big)$$

Applying the same for b, we see that $f(a)f\big(f(a)\big) + f\big(af(a)\big) = f(b)\, f\big(f(b)\big) + f\big(b\, f(b)\big)$ and $2\, f\big(f(a)\big) = 2\, f\big(f(b)\big)$. Therefore, $a = b$.

Claim: $f(x) = \frac{1}{x}$.

Note that, by (B),

$$f(x)\, f(f(x)^2) = f\big(f(x)\big) \Leftrightarrow f(f(x)^2) = \frac{f\big(f(x)\big)}{f(x)}$$

Using (A), however,

$$\frac{f\big(f(x)\big)}{f(x)} = f\big(f(x^2)\big)$$

Therefore,

$$f(f(x)^2) = f\big(f(x^2)\big)$$

$$\Leftrightarrow f(x)^2 = f(x^2)$$

Applying (B) once more,

$$\Leftrightarrow f(x)^2 = \frac{f(x)}{x}$$

$$\Leftrightarrow f(x) = \frac{1}{x}$$

Problem 22 (10SLA6)

(I didn't get close to solving this, this is pretty much the official solution.)

Let A be the smallest number in the range of f and B be that of g.

Since $f(n) + 1 = f\big(g(n)\big)$, we may induct to get the ranges of f and g are $\{A, A+1, \ldots\}$ and $\{B, B+1, \ldots\}$ respectively. WLOG, let $A \leq B$.

Plugging in $f(n)$ into the first equation,

$$g\Big(f\big(g(n)\big)\Big) = g\big(g(n)\big) + 1$$

$$\Leftrightarrow g(f(n) + 1) = g\big(g(n)\big) + 1$$

Inducting, we have,

$$g(f(n) + k) = g^{k+1}(n) + 1$$

Lemma: $g(n) \geq f(n) + (B - A)\ \forall\, n \geq 1$

We first prove that, for all k,

$$n \geq A + k \Longrightarrow g(n) \geq B + k + 1$$

We prove this by strong induction on k.

Base case: $k = 0$.

We have $n \geq A \Longrightarrow g(n) \geq B + 1$. We write $n = f(t)$ for some t,

$$g\big(f(t)\big) = g(t) + 1 \geq B + 1$$

Inductive step:

We start with the fact that,

$$g(n) \geq B;$$

Since $g(n) \geq A \Longrightarrow g\big(g(n)\big) \geq B + 1$,

$$g\big(g(n)\big) \geq B + 1;$$

Since $g\big(g(n)\big) \geq A + 1 \Longrightarrow g\Big(g\big(g(n)\big)\Big) \geq B + 2$,

$$g\Big(g\big(g(n)\big)\Big) \geq B + 2$$

Going forward, we'd eventually get,

$$g^{k+1}(n) \geq B + k$$

$$\Leftrightarrow g^{k+1}(n) + 1 \geq B + k + 1$$

$$\Leftrightarrow g(f(n) + k) \geq B + k + 1$$

This gives the desired conclusion.

Conclusion: By mathematical induction the lemma is proven.

Claim: For any integers $k \geq 1$ and $n \geq 1$, we have $f^{(k)}(n) \neq n$.

We may get by induction that, $g(n) + k = g\left(f^{(k)}(n)\right)$. However, if the claim was true, then that'd mean $g(n) + k = g(n)$. (!)

We may now show that,

$$f(n) = n + 1; g(n) = n + (B - A + 1)$$

We induct on $n \geq A$.

Base case: Pick an element x_0 such that $g(x_0) = B$. By our lemma, $B = g(x_0) \geq f(x_0) + (B - A)$. This is only possible if $f(x_0) = A$. Plugging x_0 into the second equation gives,

$$g(A) = B + 1$$

Using the lemma once again, we then get $f(A) \leq A + 1$. Since $f(A) < A$ is impossible due to the range of f and $f(A) = A$ is impossible by the claim, $f(A) = A + 1$.

Inductive step:

Assume that we have f and g up to $n-1$.

$$g(n) = g\big(f(n-1)\big) = g(n-1) + 1 = n + (B - A + 1)$$

Once again,

$$f(n) \leq n + 1$$

If $f(n) < n + 1$, we'd have a chain, since we already have,

$$f(A) = A + 1;$$

$$f(A+1) = A + 2;$$

$$\ldots$$

$$f(n-1) = n$$

Therefore, $f(n) = n + 1$.

We plug this into the first equation,

$$f(n + B - A + 1) = n + 2$$

$$B = A$$

Therefore, $f \equiv g$.

+13 hours

All problems complete. 81 clubs, 17+13 = 30 hours total.